AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 34 · STUDENT LESSON

Capstone controlled experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Match the controls. Trace both paths. Select on validation.

Compare two learning rates under the same data, starting weight and update budget.

Keep one habit: Versioned plan → full update ledger → raw validation predictions → separate paired picture → supported third path → executable audit → independent target.

MAIN LESSON · SECTION 1

Change one factor and retain the complete training paths

> A matched comparison needs more than two final scores.

Learning goals

I can trace matched two-update learning-rate experiments.

I can select a candidate using a prespecified validation rule.

I can explain the limits of a one-factor comparison.

Retrieve plans and measurement

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- What must stay fixed when comparing learning rates?

> **My starting explanation**

End goal

Explain capstone controlled experiments with a traceable toy-network case and a bounded conclusion.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Trace matched two-update learning-rate experiments

Select a candidate using a prespecified validation rule

Explain the limits of a one-factor comparison

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Version the plan before comparing two learning rates

> Keep the model, data, initial state and update budget fixed.

Complete matched experiment settings

Continue with **training** (1,2) and (2,4), validation (3,6) and (4,8), and **prediction** max(0, a*x). Compare learning rates 0.1 and 0.2. Each run starts at a = 1, has no bias and takes two full-batch updates.

Keep the loss, data order and arithmetic fixed. This toy model has no randomness; repeating the exact run checks replay, not variation across seeds.

Explicit plan revision and validation selection rule

Record the comparison as an explicit revision to the one-candidate Week 33 plan before viewing new results. Validation may guide selection; sealed final data must not.

Select lower validation MSE, with a tie going to learning rate 0.1.

Name supplied and missing facts

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Trace both updates rather than only the winning score

> The restricted gradient applies while the training ReLU stays active.

Derive the training objective and gradient

For positive weights and these positive **input**s, the ReLU is active. **Training** MSE is ((a-2)^2 + (2*a-4)^2)/2 = 2.5*(a-2)^2. Its gradient is 5*(a-2). Update a to a - rate*5*(a-2). At rate 0.1, weights are 1.5 then 1.75.

At rate 0.2, weights are 2 then 2. The formula assumes the active region; do not extend it blindly to nonpositive weights.

Complete intermediate-state ledger

Rate · Update · Weight before · Training gradient · Weight after · Validation MSE

> **Rate:** 0.1
>
> **Update:** 1
>
> **Weight before:** 1
>
> **Training gradient:** -5
>
> **Weight after:** 1.5
>
> **Validation MSE:** 3.125

> **Rate:** 0.1
>
> **Update:** 2
>
> **Weight before:** 1.5
>
> **Training gradient:** -2.5
>
> **Weight after:** 1.75
>
> **Validation MSE:** 0.78125

> **Rate:** 0.2
>
> **Update:** 1
>
> **Weight before:** 1
>
> **Training gradient:** -5
>
> **Weight after:** 2
>
> **Validation MSE:** 0

> **Rate:** 0.2
>
> **Update:** 2
>
> **Weight before:** 2
>
> **Training gradient:** 0
>
> **Weight after:** 2
>
> **Validation MSE:** 0

> **Rate:** Both
>
> **Update:** Same two training rows
>
> **Weight before:** Same two validation rows
>
> **Training gradient:** No bias
>
> **Weight after:** No random operations
>
> **Validation MSE:** Final not opened

Baseline and final candidate scores

Run	Final weight	Training MSE	Validation MSE
Baseline	1	2.5	12.5
Rate 0.1, two updates	1.75	0.15625	0.78125
Rate 0.2, two updates	2	0	0

Follow the selection lanes

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Same initial a = 1	Rate 0.1, two updates	Validation MSE 0.78125
Same initial a = 1	Rate 0.2, two updates	Validation MSE 0
Same validation rows	Choose lower MSE	Select rate 0.2

The selected run fits this exact constructed relationship. Its zero error does not establish universal reliability.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Select using the written rule and the same validation rows

> A lower validation score is development evidence.

Current task and response

- Compute both updates for rates 0.1 and 0.2 using gradient 5*(a-2).

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured confounder

- Change: Change B's **training** set as well as the feature connection.

▼ Check after predicting

Prediction: The experiment no longer isolates the connection change.

Reason: A **training**-data difference supplies an alternative explanation for the observed result.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Inspect paired outcomes and remaining limits

> Do not multiply the number of distinct cases by the number of runs.

Main raw predictions and controlled-comparison limits

At a = 1.75, validation **prediction**s are 5.25 and 7. Errors -0.75 and -1 square to 0.5625 and 1, giving MSE 0.78125. At a = 2 the **prediction**s equal 6 and 8, so MSE is zero. Rate 0.2 wins the written validation rule.

This selection uses two validation cases, not four independent cases because two runs used them.

If one run used extra epochs or different **training** rows, the comparison would mix factors. Either rerun a matched experiment or report the differences and avoid claiming an isolated learning-rate effect.

A zero validation score is also compatible with a very simple task and says nothing about a new distribution.

Separate picture, complete paired flags and scope

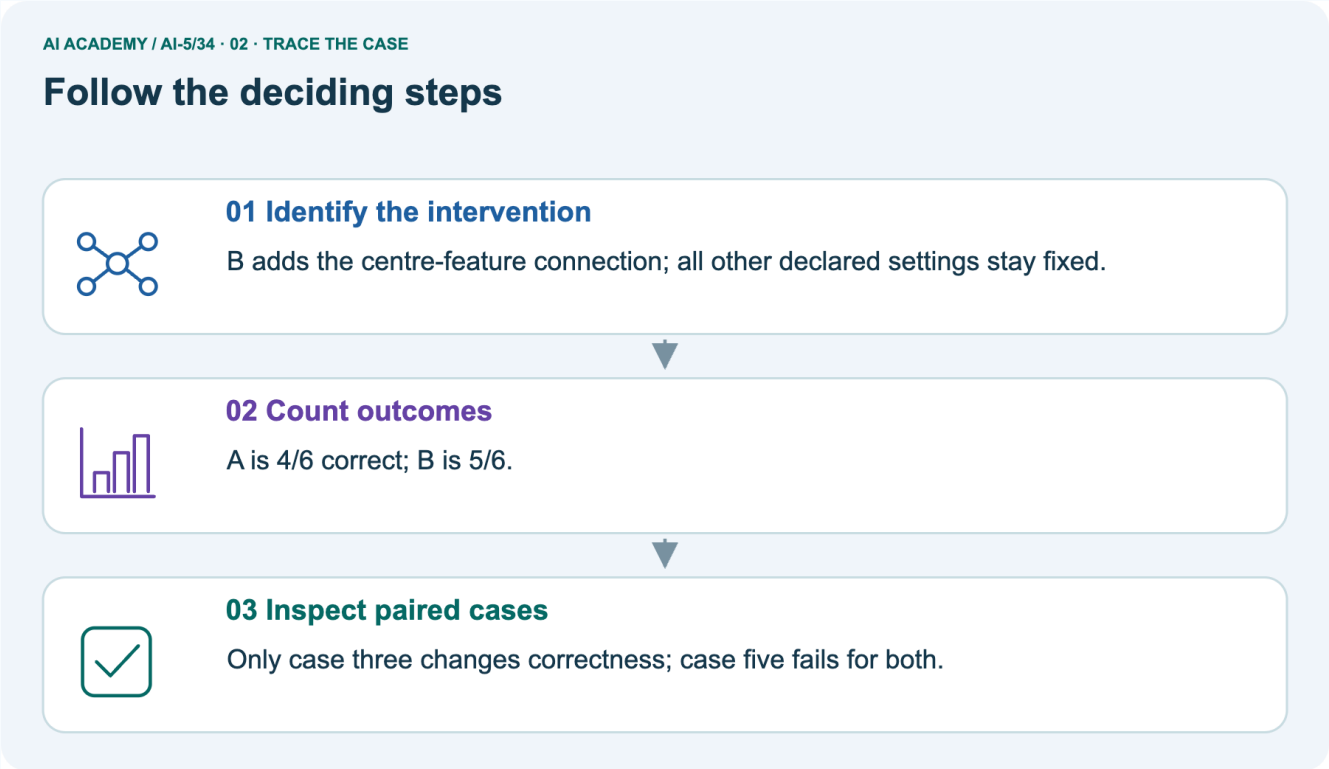
One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1]. Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

Keep the two experiments separate. The main experiment compares learning rates on scalar regression, using MSE and two validation rows. The picture supplies six paired correctness flags for two image-pattern procedures: A [1, 1, 0, 1, 0, 1], B [1, 1, 1, 1, 0, 1]. Entries are correct/incorrect flags, not predicted class labels. A has 4/6 correct and B 5/6; only case 3 changes and case 5 fails for both.

Actual pixel grids, weights and training executions are not provided, so the picture supports an audit of supplied outcomes, not a claimed model run.

Trace the supplied outcome difference



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify the intervention	B adds the centre-feature connection; all other declared settings stay fixed.
2. Count outcomes	A is 4/6 correct; B is 5/6.
3. Inspect paired cases	Only case three changes correctness; case five fails for both.

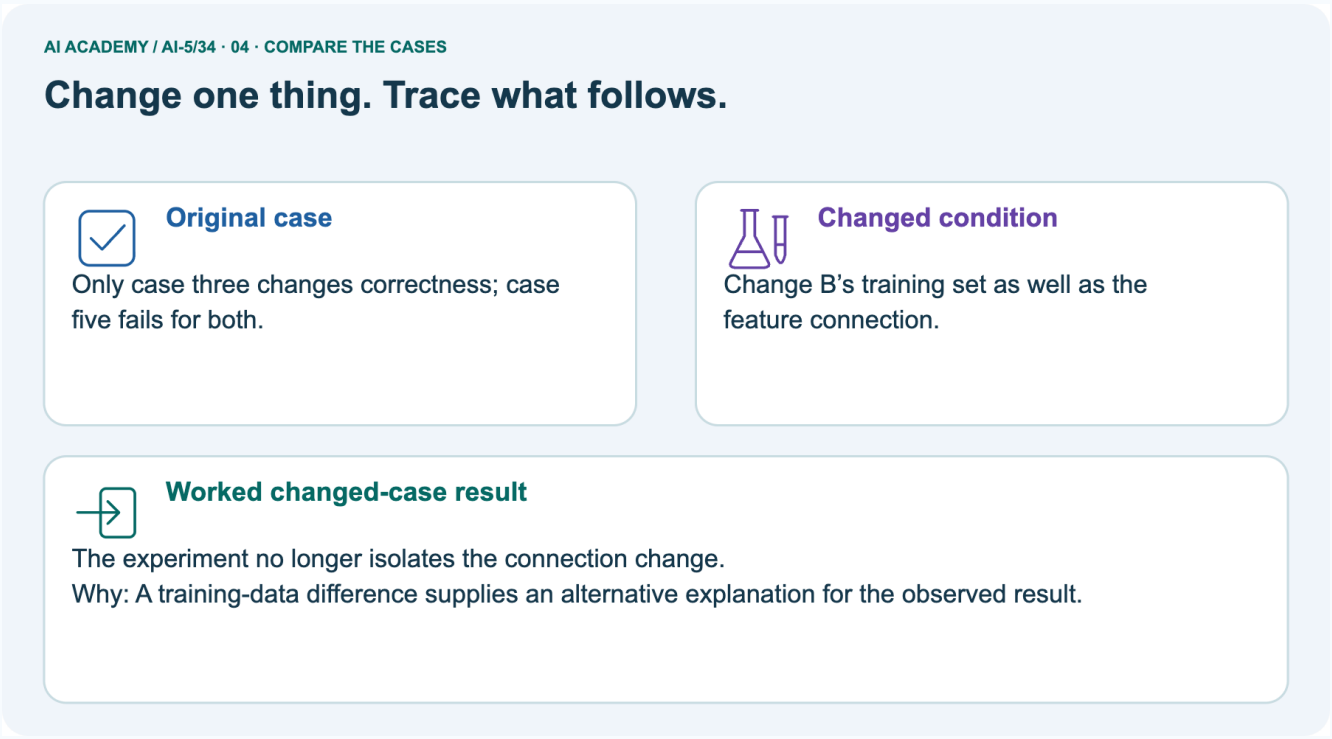
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

Predict changing both data and connection

- Change one thing: Change B's **training** set as well as the feature connection. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the changed case and limits

- These descriptive toy results do not establish significance, stable superiority or the full learned **mechanism**.

- Read the labelled picture: Compute the two correct counts and identify the changed case. Why would changing B's **training** data weaken the conclusion about its added connection?

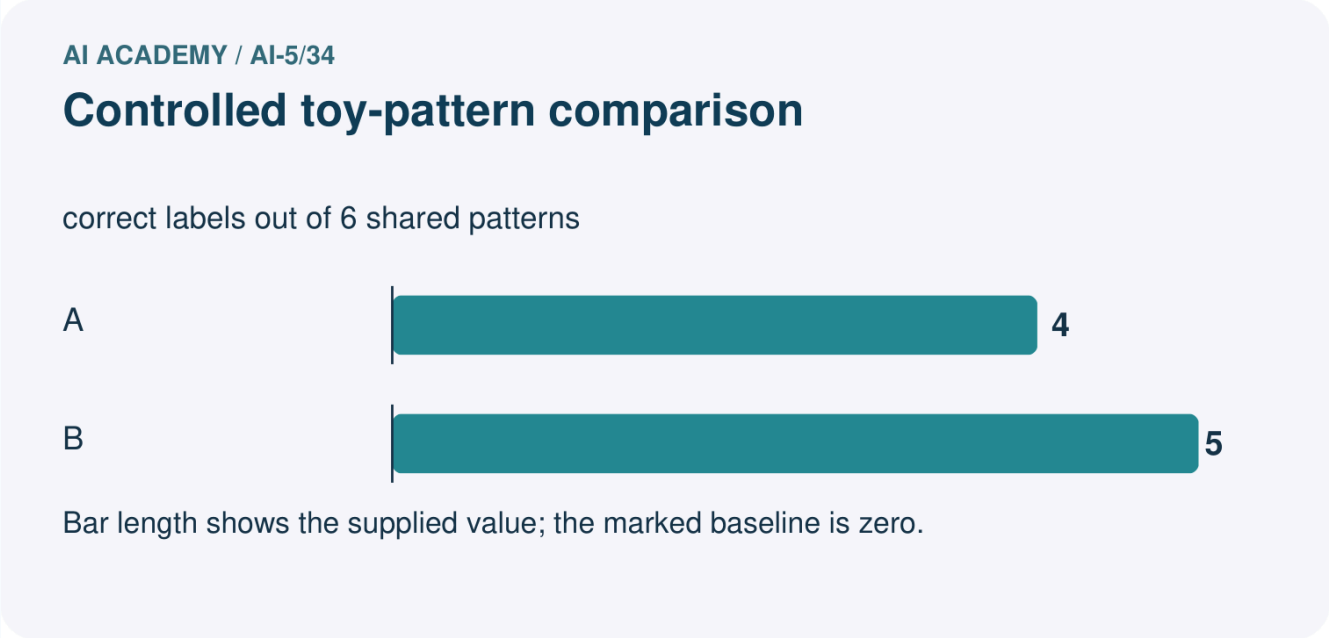
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone controlled experiments

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/34

Capstone controlled experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the intervention

B adds the centre-feature connection; all other declared settings stay fixed.



2. Count outcomes

A is 4/6 correct; B is 5/6.



3. Inspect paired cases

Only case three changes correctness; case five fails for both.

CHANGE ONE THING

Change B's training set as well as the feature connection.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Use a supported third path without changing the candidate set

> Equal final weights can follow different intermediate trajectories.

Complete separate rate 0.3 probe and guided question

- Supported example: rate 0.3 starts at a = 1. The first gradient is -5, giving a = 2.5. The next gradient is 2.5, giving a = 1.75. This two-update run finishes at the same weight as rate 0.1 although its path differs.

Guided: calculate the second update and its validation MSE. Explain why equal final scores do not mean identical trajectories.

Recompute scores and distinct-case count

- Recompute validation MSE for the final weights and apply the written **selection rule**. How many distinct validation rows were used?

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise controls, update traces and validation selection

> List fixed settings before attributing a difference to learning rate.

Current task and supported third path

- List six settings held fixed and the single intended changed factor.

> **Guided: calculate the second update and its validation MSE. Explain why equal final scores do not mean identical trajectories.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example “Capstone controlled experiments”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual experiment-reasoning check

- Use the worked case for Capstone controlled experiments. Proposed explanation: “More runs mean more validation examples”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured paired-case response

- Compute the two correct counts and identify the changed case. Why would changing B’s **training** data weaken the conclusion about its added connection?

> **My explanation using a pictured step and the given facts**

Reserve independent target y = 3x

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace both runs

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run the complete matched experiment audit

> Separate the two selection candidates, a supported probe and the supplied picture flags.

Current two-update task

Compute both updates for rates 0.1 and 0.2 using gradient 5*(a-2).

Complete executable comparison

```
"""Matched supplied regression experiments; final cases remain absent."""
train=[('T1',1,2),('T2',2,4)]
validation=[('V1',3,6),('V2',4,8)]
def predict(a,x):return max(0,a*x)
def mse(a,rows):
    if not rows:raise ValueError('Need evaluated rows')
    return sum((predict(a,x)-y)**2 for _,x,y in rows)/len(rows)
def fit(rate,updates=2):
    if rate<=0 or type(updates) is not int or updates<=0:
        raise ValueError('Need positive rate and integer update count')
    a=1.0
    history=[]
    for update in range(1,updates+1):
        if any(a*x<=0 for _,x,_ in train):
            raise ValueError('Restricted gradient needs positive preactivations')
        before=a
        gradient=sum(2*(a*x-y)*x for _,x,y in train)/len(train)
        a-=rate*gradient
        history.append((update,before,gradient,a,mse(a,train),mse(a,validation)))
    return {'rate':rate,'weight':a,'history':history,'validation_mse':mse(a,validation)}
runs=[fit(.1),fit(.2)]
selected=min(runs,key=lambda r:(r['validation_mse'],r['rate']))
assert [[r[3] for r in run['history']] for run in runs]==[[1.5,1.75],[2,2]]
assert [r['validation_mse'] for r in runs]==[.78125,0]
assert selected['rate']==.2
print('Baseline train/validation MSE:',mse(1,train),mse(1,validation))
for run in runs:
    print('Rate:',run['rate'],'history update,before,gradient,after,train MSE,valid
MSE:',run['history'])
    print('Validation ID,prediction,squared error:',
```

```
          [(name,predict(run['weight'],x),(predict(run['weight'],x)-y)**2)
           for name,x,y in validation])
print('Selected using two distinct validation rows:',selected['rate'])
# Supported .3 probe is outside the prespecified .1/.2 candidate set.
probe=fit(.3)
assert [r[3] for r in probe['history']]==[2.5,1.75]
print('Supported .3 path, not a selection candidate:',probe['history'])
# Separate supplied picture entries are correctness flags, not class predictions.
a_flags=[1,1,0,1,0,1];b_flags=[1,1,1,1,0,1]
assert sum(a_flags)==4 and sum(b_flags)==5
print('Pictured changed case IDs:',[i+1 for i,(a,b) in enumerate(zip(a_flags,b_flags)) if
a!=b])
print('Final evaluation not opened; no speed or seed-variation measurement.')
```

Expected histories, checks and honest limits

1. Predict each gradient and weight before running or tracing. Both runs use initial a = 1 and two updates on training (1, 2), (2, 4), then the same validation (3, 6), (4, 8). Rate .1 ends at 1.75 with validation MSE .78125; rate .2 ends at 2 with MSE 0. Select .2 using the declared .1/.2 candidate set. The supported .3 path ends at 1.75 too but is not an extra candidate retroactively added to that selection.

   Preserve all runs, including unsuccessful ones. No random seed variation, runtime or final-test outcome is measured. The complete program checks the active training region before using its restricted gradient.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare two rates for a new target

> Use the fresh gradient and preserve the two-update budget.

Complete independent training and validation case

For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1. Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Run a new comparison independently

Attempt this without hints

For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1. Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve
</details>

MAIN LESSON · SECTION 11

Preserve both runs and the plan revision

> Save fixed settings, changed factor, histories and validation choice.

Project task and artifact

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1.

Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a controlled comparison and bounded choice

> A zero validation MSE on a tiny linear task does not establish general superiority.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Trace matched two-update learning-rate experiments

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Controlled comparison	A comparison holding specified conditions constant while changing an intended factor
Full batch	An update whose gradient uses all training rows
Selection rule	A rule for choosing a candidate using development evidence
Confounder	A differing condition that prevents isolating the intended explanation

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- List six settings held fixed and the single intended changed factor.
- Recompute validation MSE for the final weights and apply the written **selection rule**. How many distinct validation rows were used?
- Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different **training** rows. Can the difference be attributed to rate alone? Propose a repair.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent y = 3x comparison for the first individual attempt.

Current explanation and response

- Save run IDs, settings, each checkpoint, **prediction**s and metric calculations. Report the losing run as well as the winner. Freeze the selected rate, architecture and update count before the final evaluation.

  Any change after opening final data requires a new development record and fresh reserved **evidence**.

- Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different **training** rows. Can the difference be attributed to rate alone? Propose a repair.

> **Exit: one run receives six updates and the other two. What can the score comparison establish, and what must be rerun to isolate the learning rate?**

> **My limited claim and a useful next check**

▼ Extra practice · Revisit the complete trace

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Same initial a = 1	Rate 0.1, two updates	Validation MSE 0.78125
Same initial a = 1	Rate 0.2, two updates	Validation MSE 0
Same validation rows	Choose lower MSE	Select rate 0.2

The selected run fits this exact constructed relationship. Its zero error does not establish universal reliability.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

By the end, I can	How I will show it
Trace matched two-update learning-rate experiments	Explain the deciding step, show a trace or test, and state one limit.
Select a candidate using a prespecified validation rule	Explain the deciding step, show a trace or test, and state one limit.
Explain the limits of a one-factor comparison	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.
- Which labelled step matches this detail? Only case three changes correctness; case five fails for both.
- 1. Inspect paired cases
- 2. Identify the intervention
- 3. Count outcomes
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Inspect paired cases Only case three changes correctness; case five fails for both.

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

Changed-case question and optional hint

- Changed-case practice: Change B's **training** set as well as the feature connection. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The experiment no longer isolates the connection change.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The experiment no longer isolates the connection change. A **training**-data difference supplies an alternative explanation for the observed result.

Repair and retain the record

Review Capstone hypothesis and baseline (ai-5/33). Goal: Write a testable capstone hypothesis and a credible baseline. Short practice: Name the intended change, the baseline and the measured outcome.

Why must both versions avoid **training** on the eight reserved labels?. Return to this week and explain what you changed.

Review Efficiency and resource measurement (ai-5/31). Goal: Calculate throughput and amortised time per example from repeated timings. Short practice: Why is 10 ms/case not necessarily a single-request latency?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted **output** head. Short practice: Which part was trained for the new task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

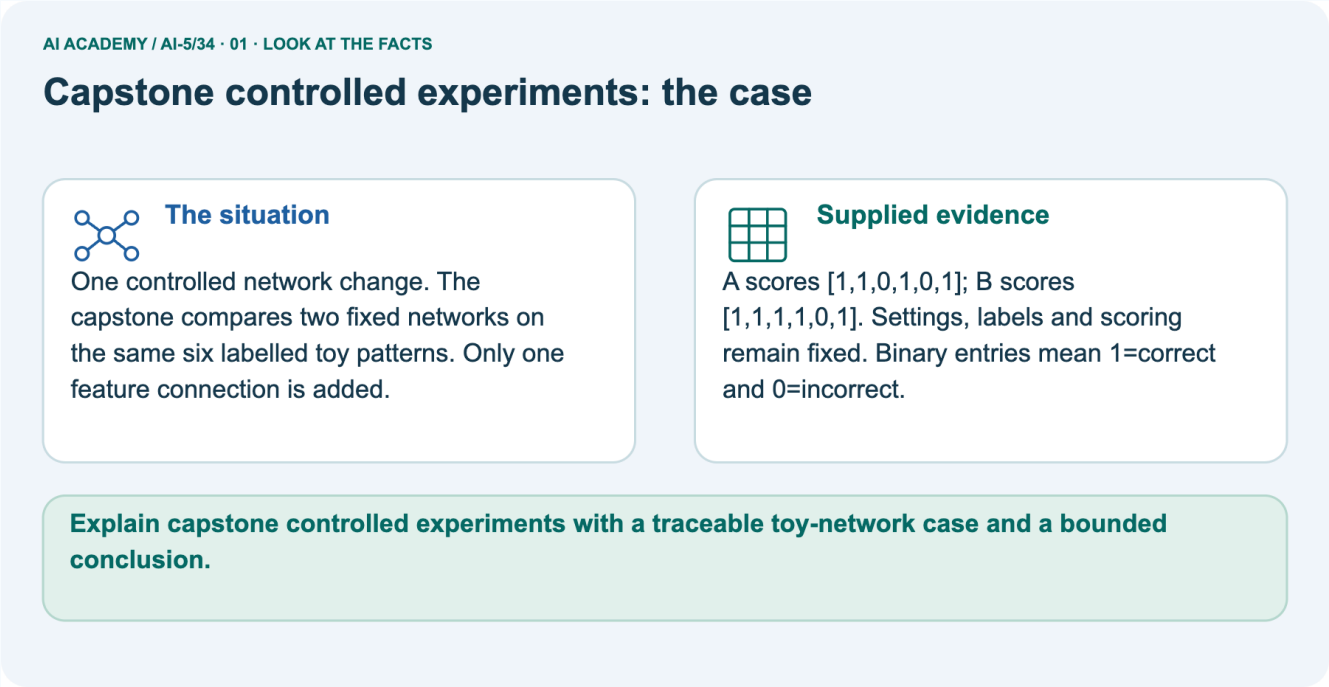
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Capstone controlled experiments: the case



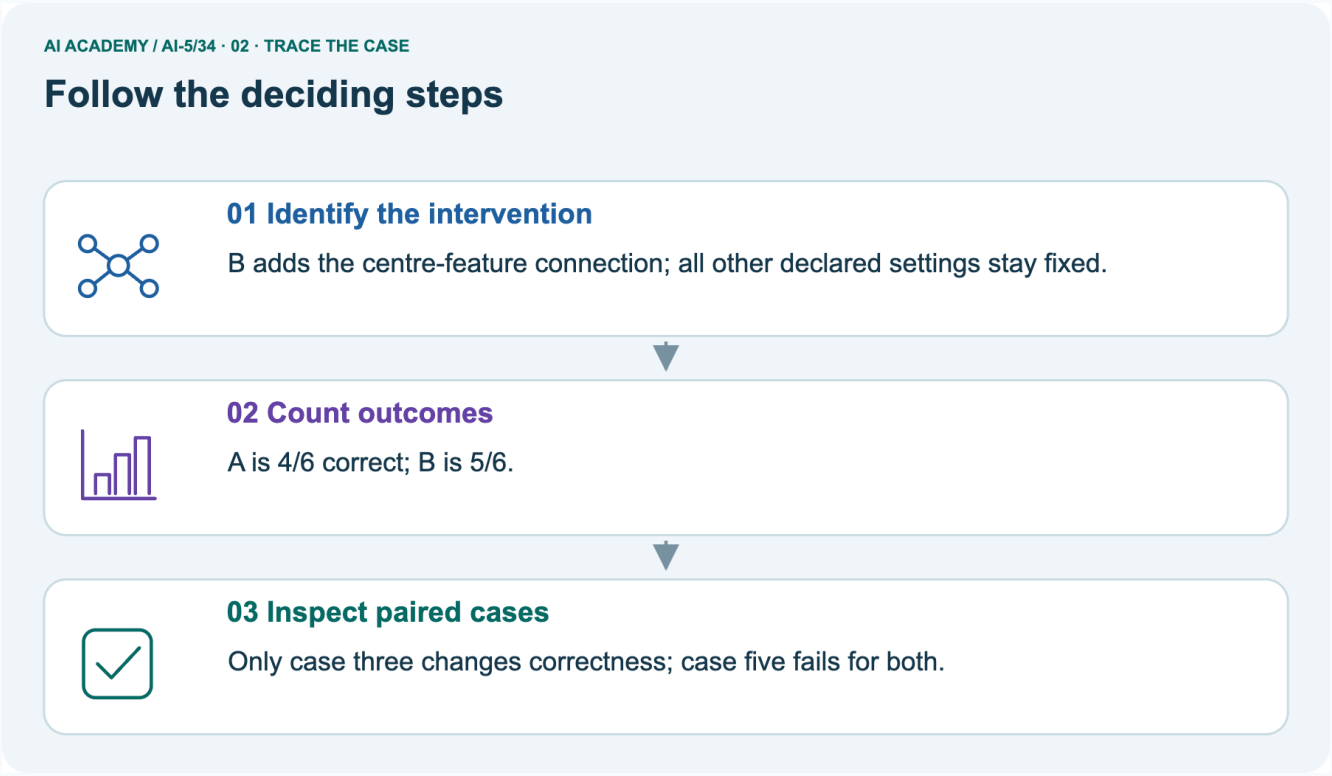
Follow the labelled picture with the matching explanation. Open the vector diagram

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added. Supplied facts: A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1].

Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

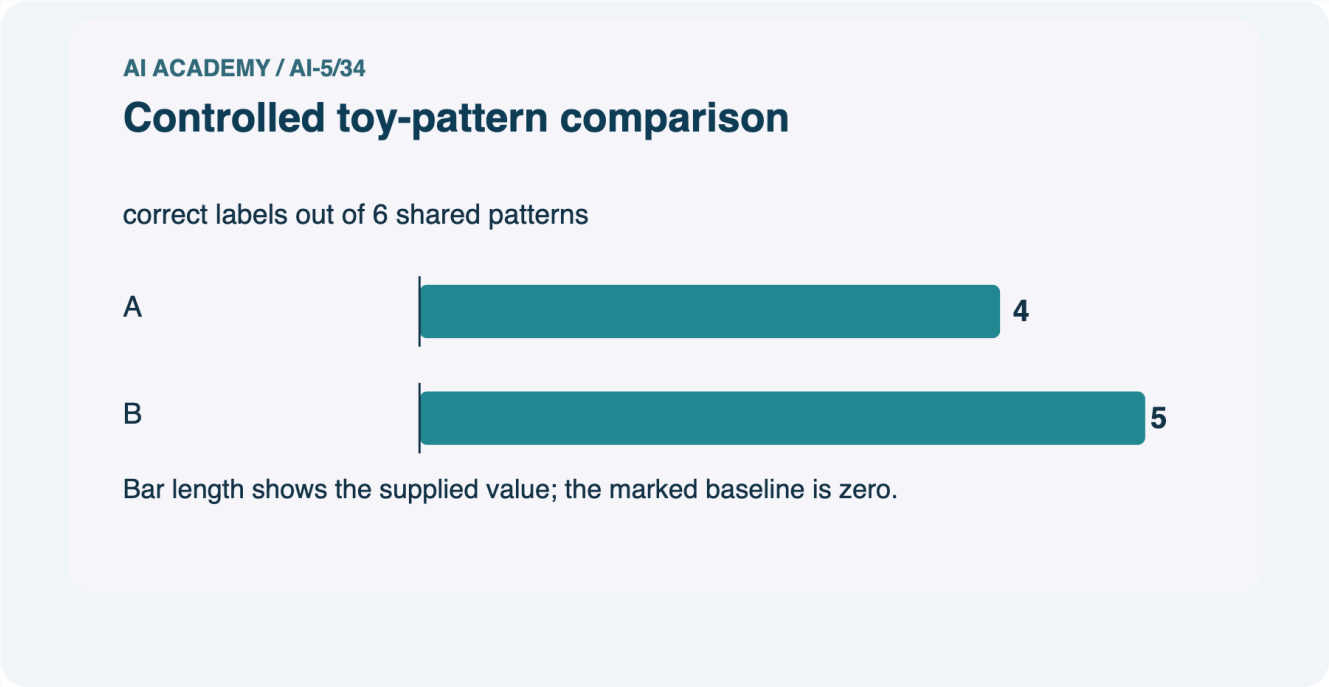


Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the intervention: B adds the centre-feature connection; all other declared settings stay fixed. → Count outcomes: A is 4/6 correct; B is 5/6. → Inspect paired cases: Only case three changes correctness; case five fails for both.

Point to each numbered stage and explain which recorded **input** it uses.

03. Controlled toy-pattern comparison

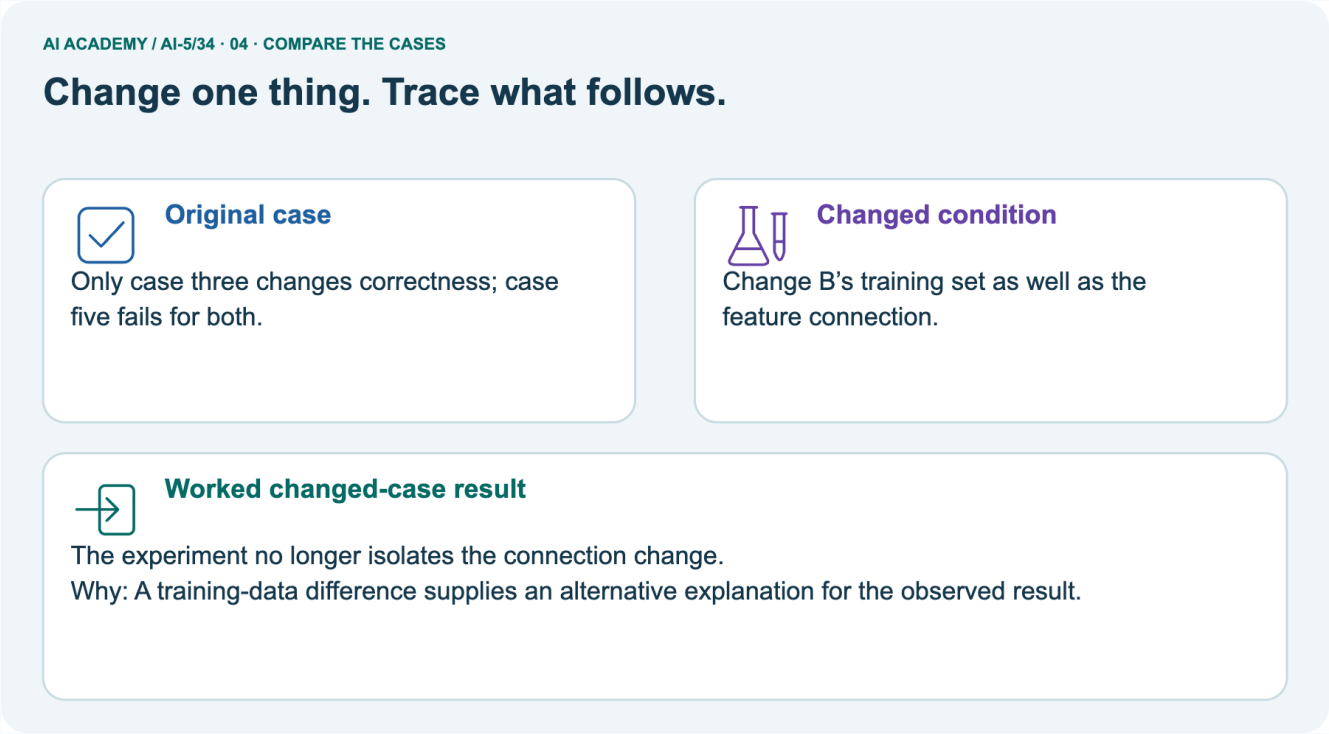


Follow the labelled picture with the matching explanation. Open the vector diagram

Controlled toy-pattern comparison. Units: correct labels out of 6 shared patterns. A: 4. B: 5.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



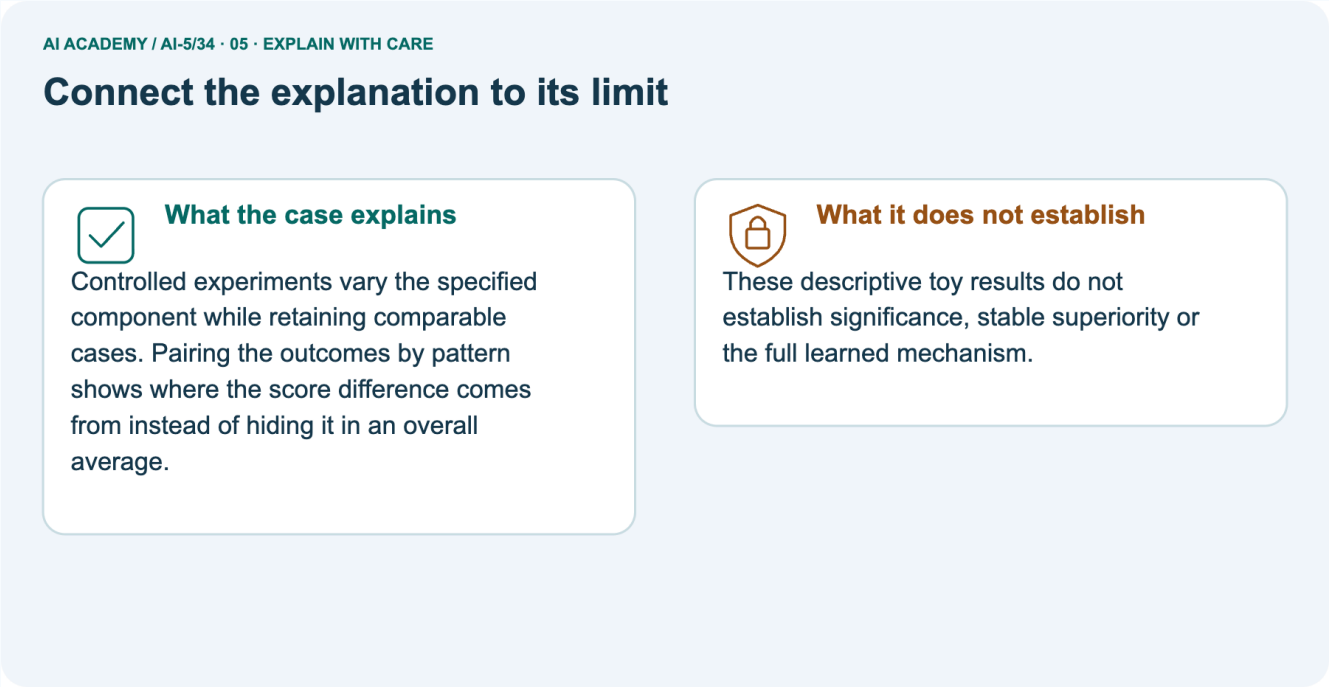
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only case three changes correctness; case five fails for both. Changed condition: Change B's **training** set as well as the feature connection.

Expected result: The experiment no longer isolates the connection change. Reason: A **training**-data difference supplies an alternative explanation for the observed result.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



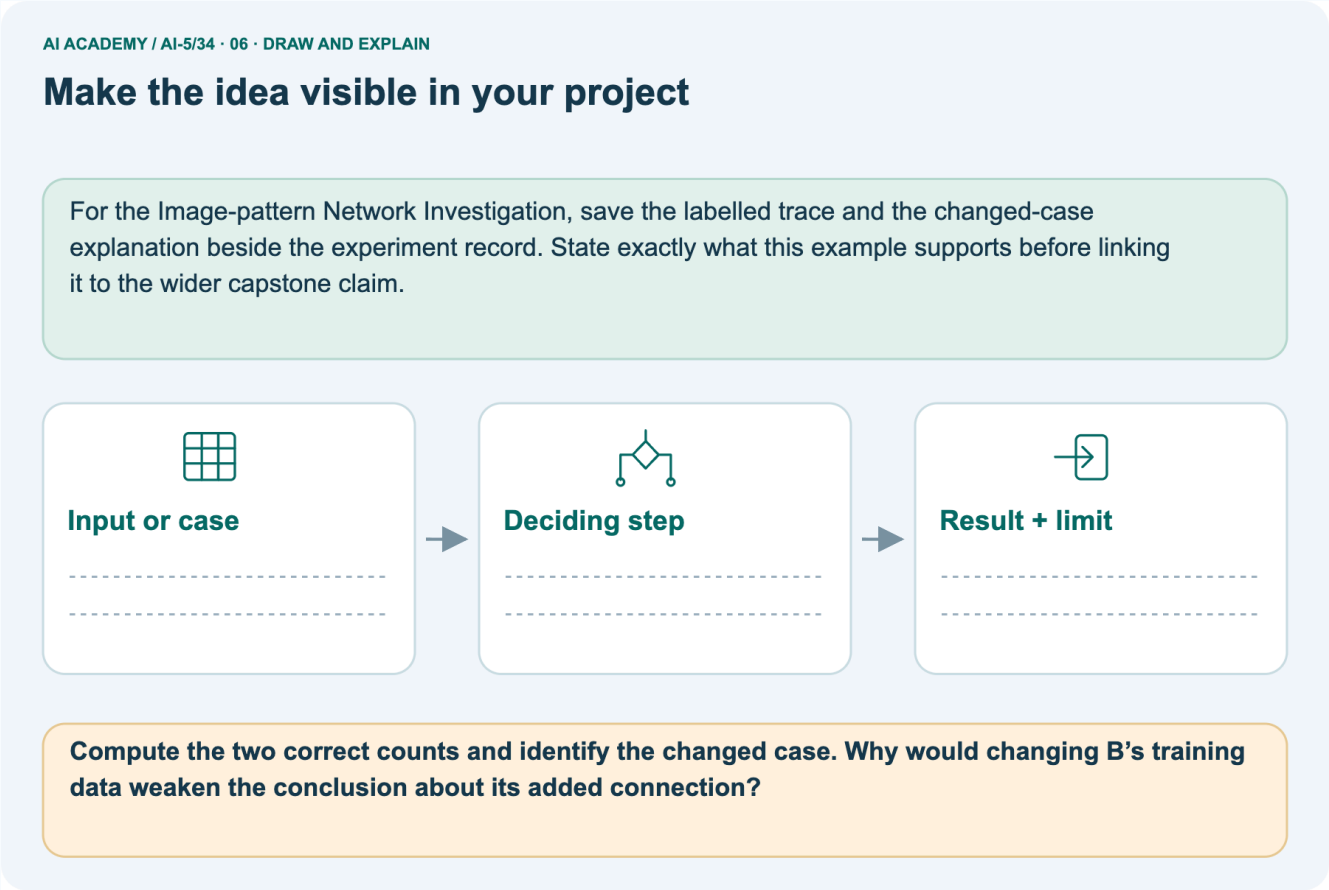
Follow the labelled picture with the matching explanation. Open the vector diagram

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

Limit: These descriptive toy results do not establish significance, stable superiority or the full learned **mechanism**.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Compute the two correct counts and identify the changed case. Why would changing B's **training** data weaken the conclusion about its added connection?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Capstone controlled experiments: the case

02. Follow the deciding steps

03. Controlled toy-pattern comparison

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.